



Q1

Comparison of key hardware

	ARCHER2	Bede
Nodes	5,860	32
Processor	2× AMD EPYCTM 7742, 2.25 GHz, 64-core	2x, IBM Power9, 2.7 GHz, 16-core
Cores per node	128 (2 x 64-core processor)	32 (2x 16-core processor)
Memory per node	256 GiB (standard memory), 512 GiB (high memory)	512GB DDR4 RAM
Interconnect	HPE Cray Slingshot, 2× 100 Gbps bi-directional per node	NVLink 2.0
GPU per node	Only 4 nodes with 4x AMD Instinct MI210	4x NVIDIA Tesla V100 32GB NVLink 2.0 GPUs (2 per CPU)

It is clear from the table that ARCHER2 is focused on a CPU architecture, with 750,080 cores in total. ARCHER2 has an estimated peak performance of 28 PFLOPs. However, Bede is focused on a GPU architecture, boasting many more high powered GPUs than ARCHER2.

Comments on hardware

- ARCHER2 is designed for general purpose high performance computing, running programs such as CASTEP and HadGEM3. The use of 64 core processors and a large amount of nodes means that ARCHER2 offers extreme benefits from parallel programming. On the other hand, Bede is designed for more specialised applications that need GPU resources such as machine learning, a more modern demand.

- For large parallel programs, the communication between different pieces of hardware is extremely important. ARCHER2 has a 100 Gbps interconnect which is quite fast and does help minimise communication time. However Bede's NVLink 2.0 offers a 25 Gigatransfers per second, which is an estimated 1600 Gbps. This much higher interconnect is needed for the high memory usage that GPU programs typically need.